

SYSTEMS

Algorithms to Antenna: Benchmarking Radar Cross Section

Following up our recent blog on methods that tackle radar cross section of simple point targets and extended targets, this edition zeros in on benchmarks for RCS.

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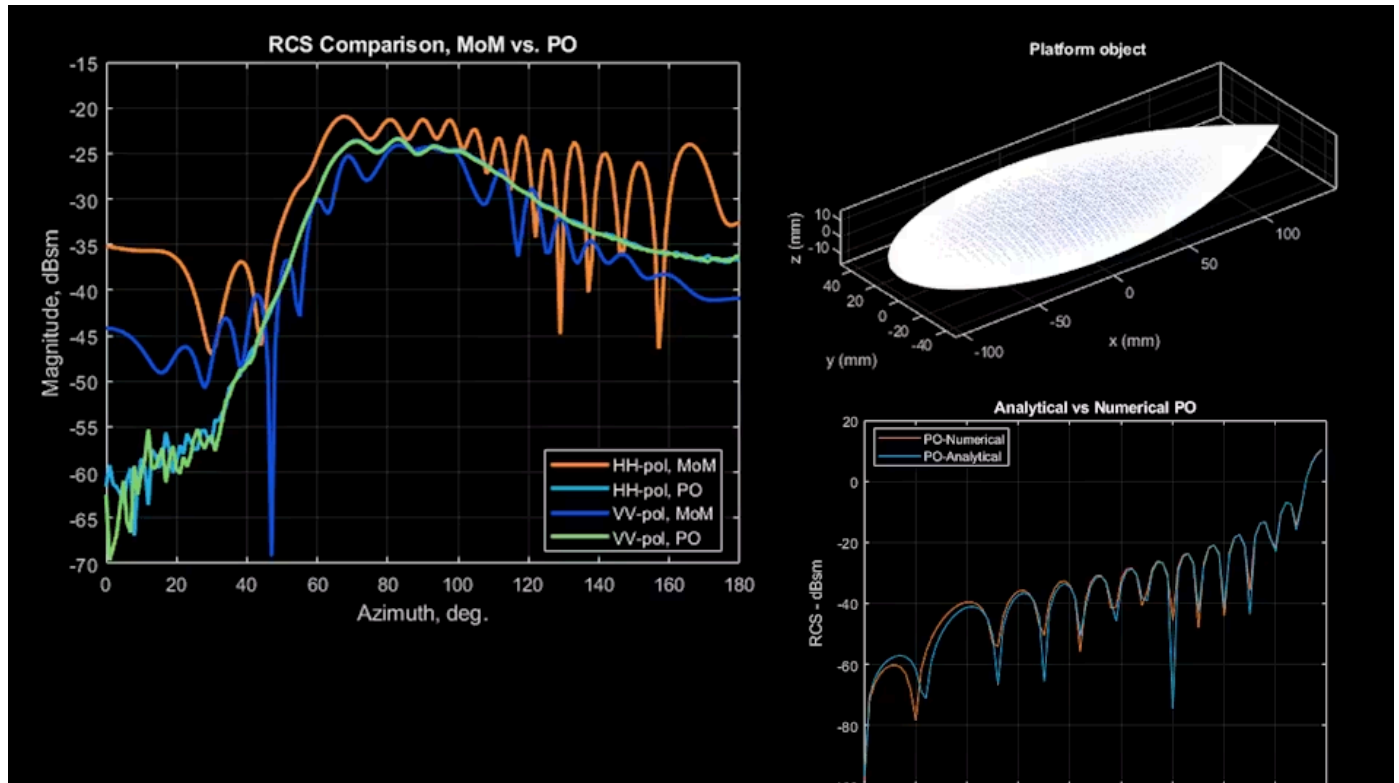
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In our [blog on radar target modeling](#), we reviewed techniques for radar cross section (RCS) of simple point targets and extended targets with multiple scattering centers. We also noted we would review computational electromagnetic (EM) approaches that can be used for more accurate RCS predictions. That brings us to our next topic: benchmarking radar cross sections.

We start with an example that benchmarks two popular structures: a square plate, a circular plate, and a circular disk. For the square and circular plates, we compare the MoM solution with the PO solution. For the circular disk, we compare the MoM solution with the PO solution. For the NAS, we compare the MoM solution with the PO solution.

Square Plate RCS

You can start with the physical dimensions of the square plate, which are defined by the side length a and the thickness t . The square plate is modeled as a perfectly conducting surface.

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```
L = 10.16e-2;
```

```
W = 10.16e-2;
```

```
p = platform;
```

```
p.FileName = 'square_plate.stl';
```

```
p.Units = 'm';
```

The RCS computation is done in the 0 degrees. The electric-field polarization on transmit and horizontal component for the RCS calculation. The results of the Toolbox are compared with the analytical textbook, *Radar System Analysis and Design* by Mahafza.

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```

az = 0;

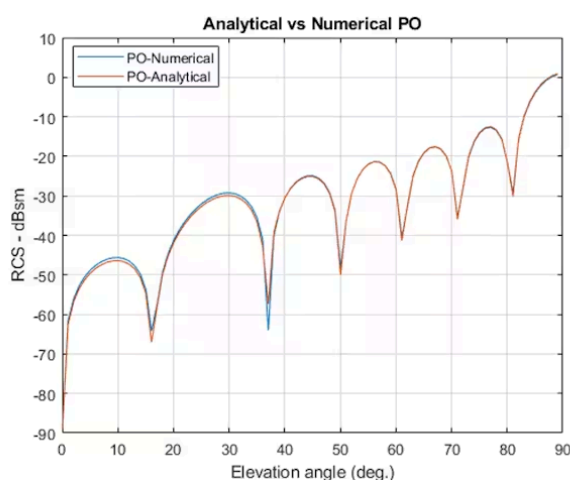
el = 0.05:1:90;

sigma = rcs(p,f1,az,el,'Polarization','HH');

asigma1 = rectPlateRCS(L,W,f1,az,90-el);

```

Figure 1 shows that the results compare closely, which is expected given the simple nature of the structure.



1. RCS for a square plate is depicted, revealing analytical results versus the numerical calculation method.

Circular Plate RCS

Let's take a look at the same plot for a circular plate. Again, you can set the physical dimensions of the circular plate. It's described using an STL file as

follows:

```

R = 10.16e-2;

pc = platform;

pc.FileName = 'circular_pla

pc.Units = 'm';

```

Again, the results from the RCS fun

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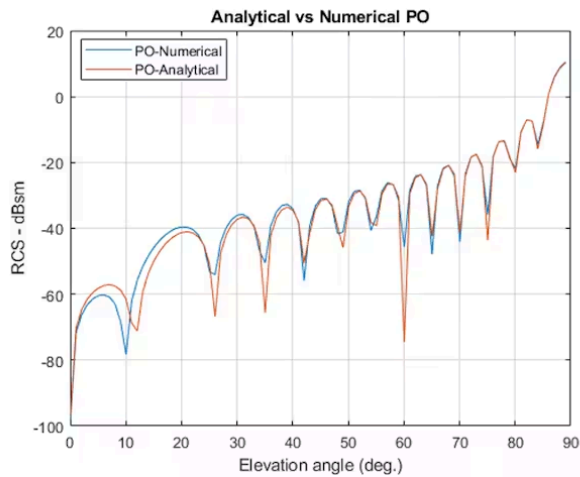
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```
e1 = 0.05:1:90;

sigmaV = rcs(pc,f1,az,e1,'Polarization','HH');

asigma1 = circPlateRCS(R,f1,90-e1);
```

Figure 2 shows that the results again compare closely.



2. Here, RCS for a circular plate is reveals the analytical result versus the numerical calculation method.

NASA Almond

The third structure we look at is the NASA almond shape, which is also an important shape for benchmarking the performance of high-frequency EM

solvers. The shape produces a large dynamic scattered field over large angular regions. The mathematical expressions in [IEEE's EM programmer's notebook—benchmark radar targets for the validation of computational electromagnetics programs reference](#) have been used to create the STL file that describes the almond shape. Again, this structure can be brought into MATLAB as follows:

```
p = platform;

p.FileName = 'NASA-Almond.s

p.Units = 'm';
```

Figure 3 shows the visualization of solver is only applicable at large **ka**

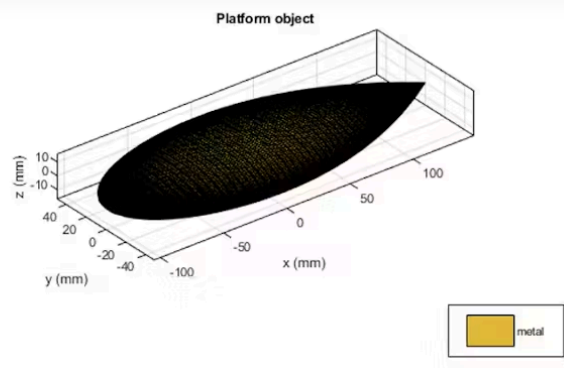
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The results are produced by two solvers: the physical optics (PO) and the MoM. For this example, the wavelength for 7 GHz is approximately 4.3 cm. We refine the mesh used by the solver to be slightly finer than this using the $\lambda/10$. The mesh is defined in MATLAB as

follows:

```
f2 = 7e9;
```

```
m = mesh(p, 'MaxEdgeLength', .0035)
```

```
az = 0:1:180;
```

```
el =0;
```

```
m =
```

```
struct with fields:
```

```
NumTriangles: 7878
```

```
NumTetrahedra: 0
```

```
NumBasis: []
```

```
MaxEdgeLength: 0.0035
```

```
MeshMode: 'manual'
```

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```

        'EnableGPU', false, ...

        'Polarization', 'HH');

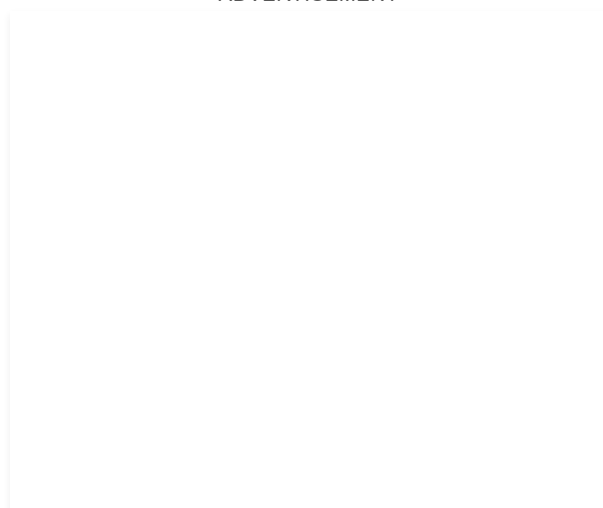
sigmahh_mom = rcs(p, f2, az, el, 'Solver', 'MoM', ...

        'Polarization', 'HH');

```

Similarly, we calculate the RCS for the VV-polarization condition, where the transmitted and received fields are both vertically polarized.

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```

sigmavv_po = rcs(p, f2, az, el, 'Solver', 'PO', ...

        'EnableGPU',

        'Polarization

sigmavv_mom = rcs(p, f2, az, e

        'Polarization

```

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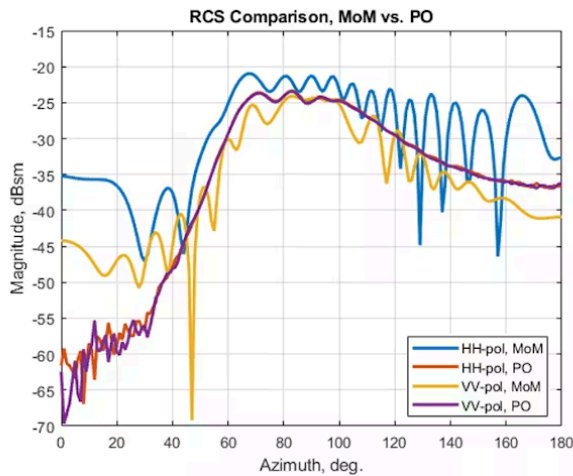
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approximation and therefore is only able to predict the level of RCS. All variations at the different angles are averaged out. This is expected since the PO solver assumes that the surface current density outside the illuminated region does not contribute to the scattered field.



4. This RCS comparison of the NASA almond shows MoM and PO methods.

You have multiple techniques to model RCS in MATLAB. You can trade off the levels of fidelity of your model to match the requirements of your program.

To learn more about the topics covered in this blog, see the links below or email

me at: rgentile@mathworks.com.

- [Modeling Target Radar Cross Section](#) (example): Learn how to model radar targets with increasing levels of fidelity.
- [Radar Cross Section Benchmarking](#) (example): Learn about RCS benchmarking.

See [additional 5G, radar, and EW resources](#), including those referenced in previous blog posts.



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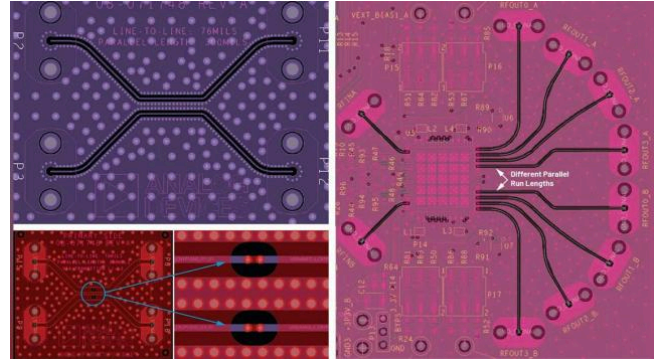
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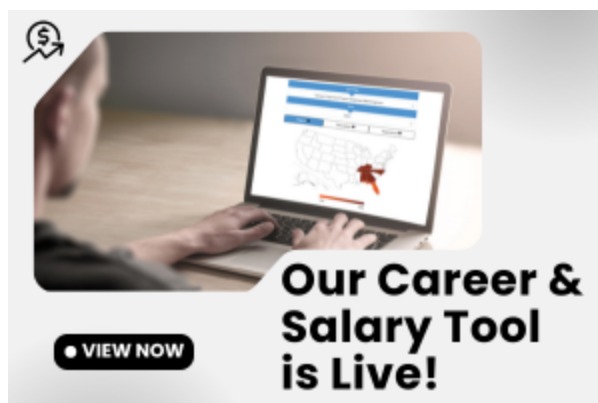
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